HAROS 2026

Health Monitoring and Retrofitting of Existing Structures

Comparative Evaluation of Unsupervised Anomaly Detection Algorithms for Structural Health Monitoring Using Real-World Bridge Sensor Data

Amelia Knapik & Marcello Balduccini

Saint Joseph's University

HAROS 2026

SJU Saint Joseph's University

Third international conference on Health Monitoring and Retrofitting of Existing Structures

AGENDA

01 Research motivation and question

02 Bridge data and analysis pipeline

03 Six unsupervised algorithms

04 Results: alignment, runtime and model agreement

05 Video context, conclusions and next steps

One hour of monitoring produced approximately 1.84 million raw sensor measurements.

- **Inspection is intermittent**

  Traditional inspection captures only snapshots of structural behavior.

- **Screening must be practical**

  Useful methods should surface unusual responses without excessive computation.

- **Ground truth is limited**

  Confirmed examples of damage are rare and require expert validation.

- **Extreme conditions may be missing from training data**

  Supervised models can become less reliable when weather or operating conditions differ from past labeled examples.



Research question: Which method best balances alignment with sensor-derived events, runtime and screening utility?

Anomaly flags prioritize engineering review. They do not diagnose structural damage.

SYNCHRONIZED SENSOR AND VIDEO WINDOWS

4

sensors analyzed: 1, 2, 4 and 5

128 Hz

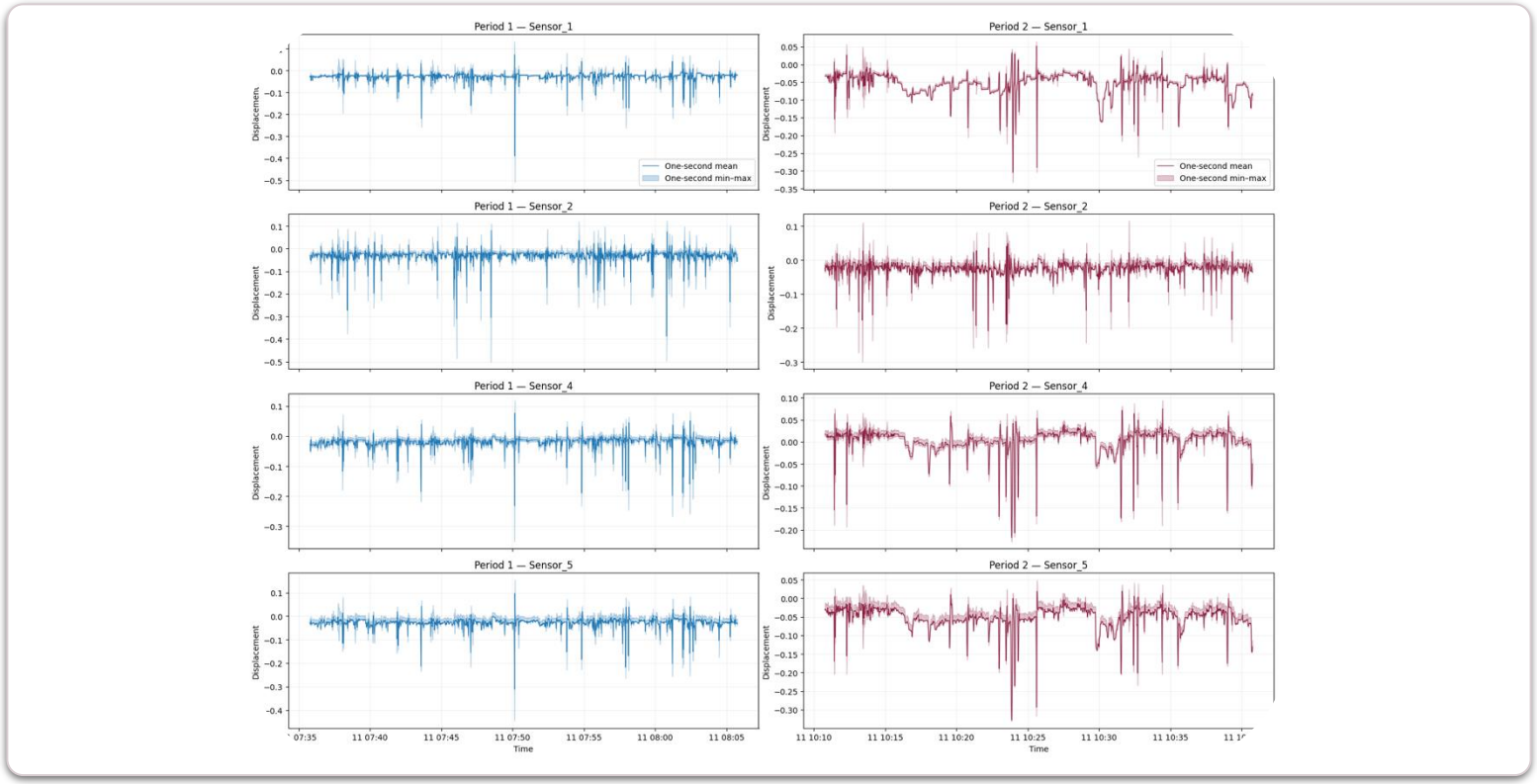
raw sampling rate

2 × 30 min

analysis windows

3,600

one-second observations



Period 1

07:35:47–08:05:46

Period 2

10:10:47–10:40:46

Video coverage: 07:35:47–12:20:48

One-day proof of concept

COMMON ANALYSIS PIPELINE

1

Clean

Remove the metadata row when present

2

Aggregate

Convert 128 readings per second into one observation

3

Create features

Calculate five summary statistics for each sensor

4

Scale features

Use RobustScaler within each period

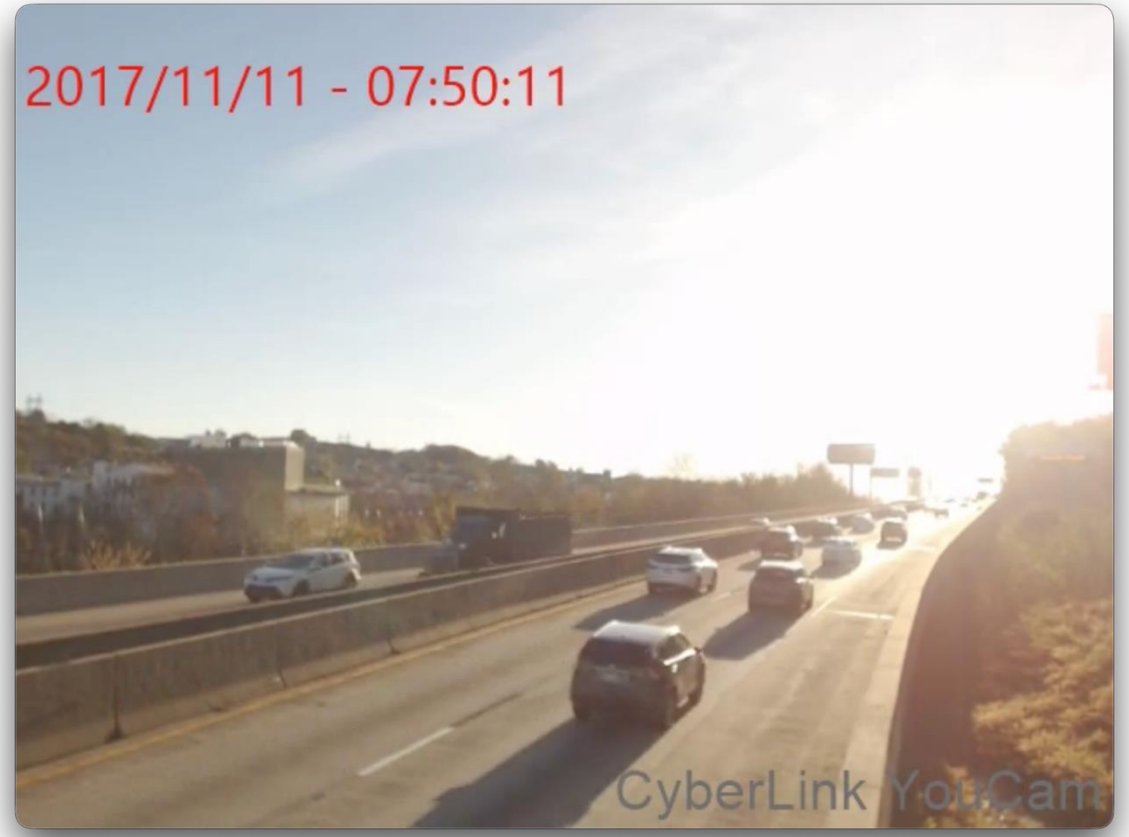
5

Compare models

Use the same 3,600 × 20 data matrix and 1% anomaly proportion

Controlled comparison: identical input features and anomaly proportion for every model

EVALUATION FRAMEWORK WITHOUT DAMAGE LABELS



20 candidate timestamps reviewed

- **Create reference events**

  Select the 10 strongest multi-sensor activity peaks per period, separated by at least 10 seconds.

- **Review physical context**

  Inspect ±10 seconds of video and annotate vehicle type, count and direction.

- **Measure model alignment**

  Count a model match when a flagged second occurs within ±5 seconds of a reference event.

Reference events come from sensor activity. Alignment is descriptive and does not measure accuracy or damage detection.

SIX UNSUPERVISED ALGORITHMS COMPARED

Ensemble

Isolation Forest

Random partitioning isolates unusual observations

Clustering

CBLOF

Cluster size and distance define outlier scores

Distance

Sampling

Subsample-based proximity identifies sparse observations

Rotation-based

ROD

Robust geometric scoring across rotated views

Neural network

Autoencoder

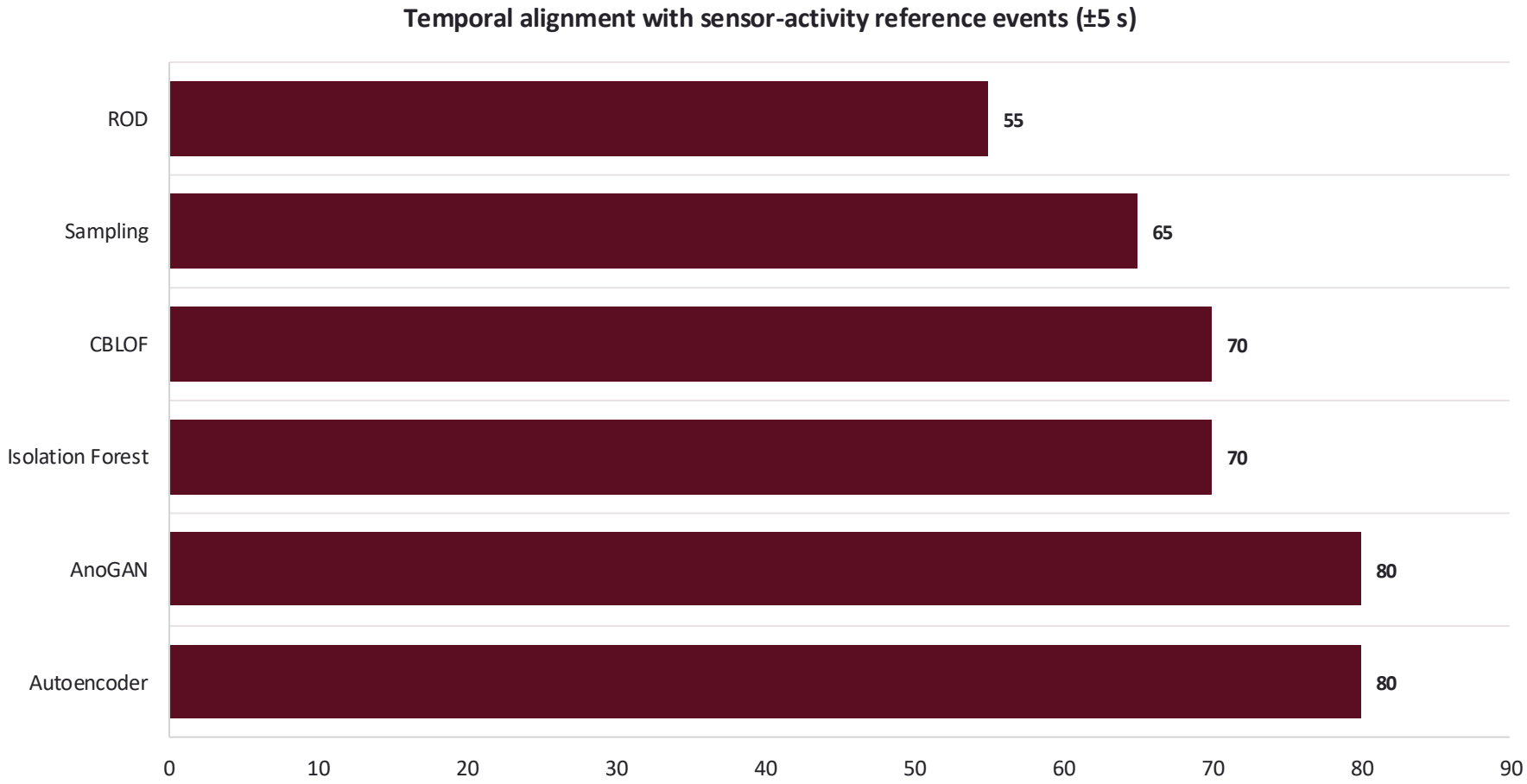
Reconstruction error measures deviation

Generative

AnoGAN

Adversarial reconstruction produces anomaly scores

AUTOENCODER AND ANOGAN SHOWED THE HIGHEST OBSERVED ALIGNMENT



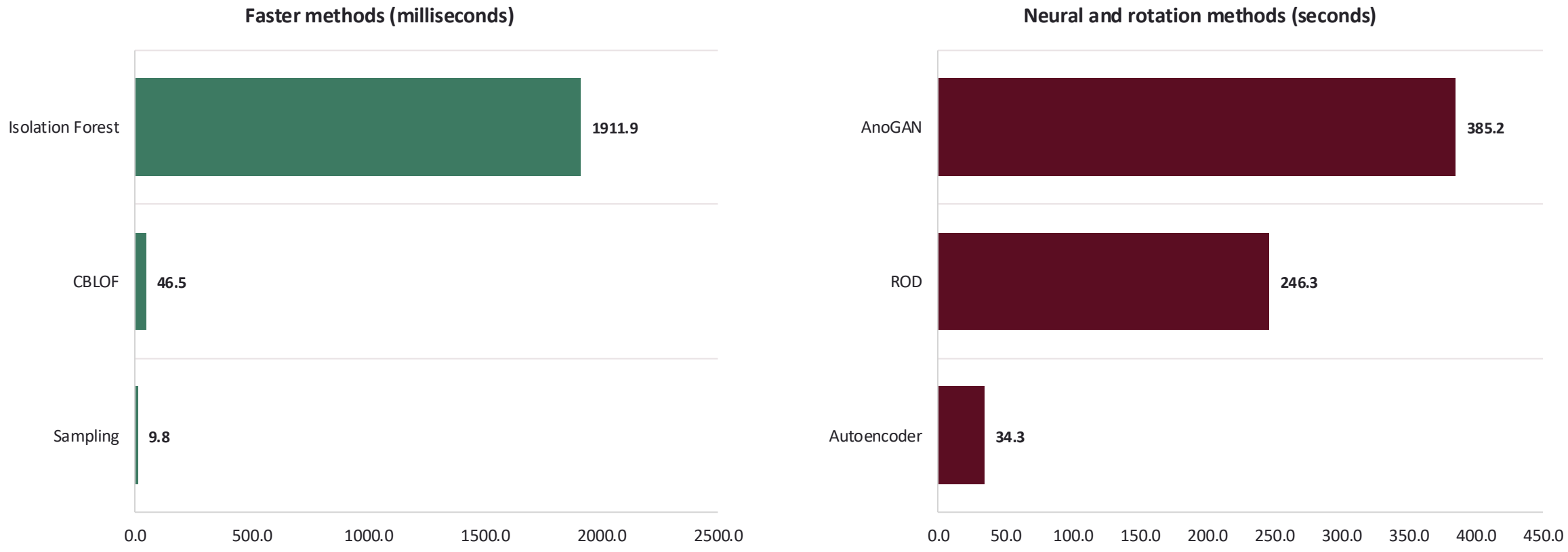
16/20

combined across both periods

Alignment with sensor-derived events. Independent accuracy and damage detection remain unmeasured.

Autoencoder 80%, AnoGAN 80%, Isolation Forest 70%, CBLOF 70%, Sampling 65%, ROD 55%

RUNTIME VARIED BY NEARLY FIVE ORDERS OF MAGNITUDE



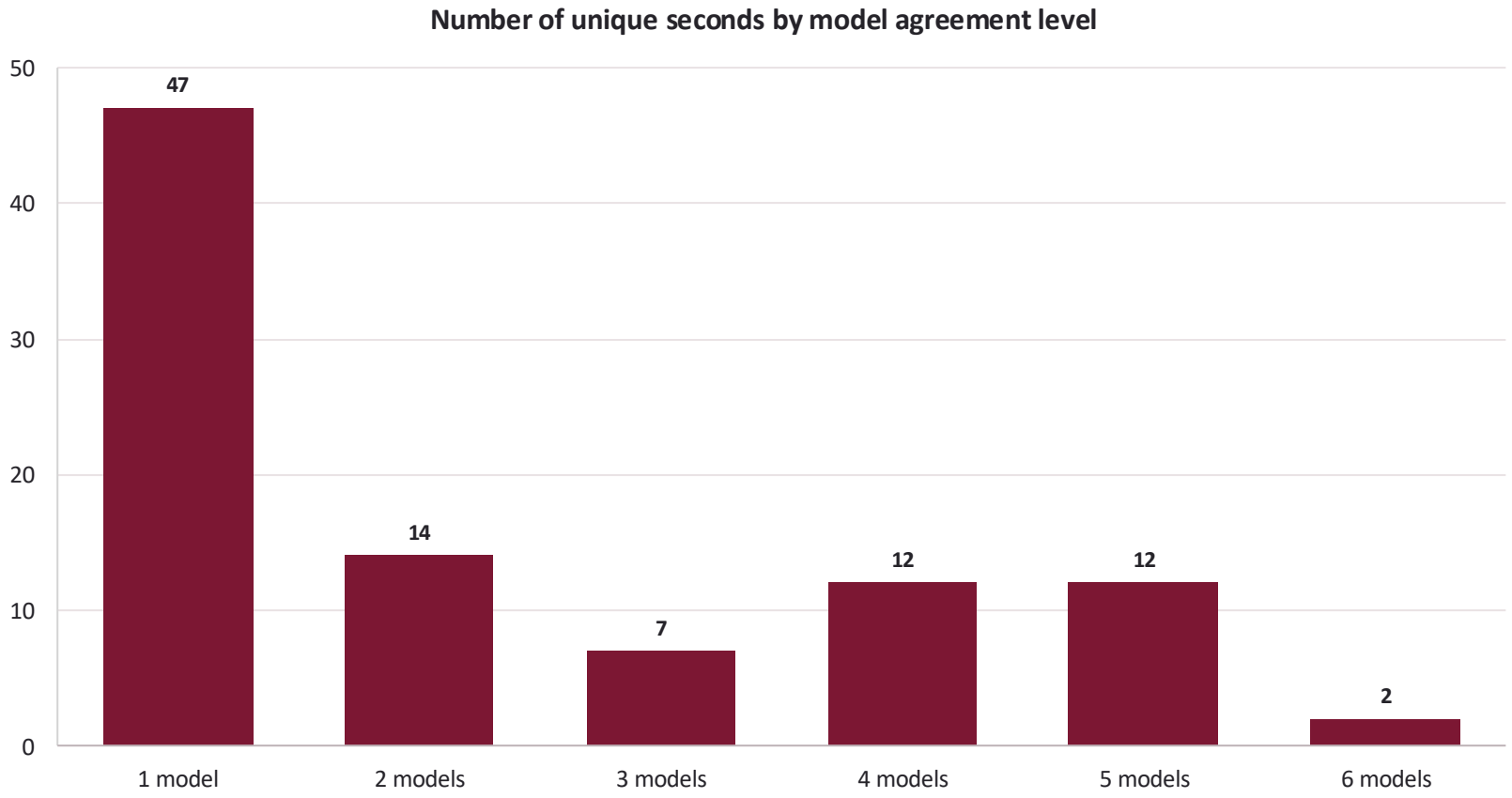
Different units: milliseconds at left, seconds at right

Sampling: 9.8 ms

AnoGAN: 6.42 min

Google Colab CPU-only, Intel Xeon 2.20 GHz, 1 physical and 2 logical cores, 12.67 GB RAM

MOST FLAGGED SECONDS WERE MODEL-SPECIFIC



94

unique seconds flagged by ≥1 model

2

seconds flagged by all six models

All-six timestamps

07:50:10
10:13:26

Each model flagged 36 seconds. ROD showed the lowest overlap with the other methods.

VIDEO CONTEXT FOR REPRESENTATIVE SENSOR EVENTS



Candidate 07:43:36 (frame +1 s)

Two heavy trucks visible within the ±10-second review window



Candidate 10:11:27 (frame +2 s)

Tanker truck visible within the ±10-second review window

Across all 20 video reviews: 19 were annotated as strong heavy-vehicle associations and 1 as a possible association.

What the current experiment supports

- **No definitive winner**

  Autoencoder had the highest observed alignment at moderate runtime. Isolation Forest was much faster with slightly lower alignment.

- **Speed and alignment must be considered together**

  Sampling was fastest. AnoGAN matched the top alignment but required the longest runtime.

- **Algorithms prioritize different seconds**

  Low complete agreement shows the value of comparing rankings, not only fixed flag counts.

Validation required next

1. Repeat runtime tests and report median ± spread
2. Review model-selected events beyond the initial 20 candidates
3. Test additional days and quantify weather-related variability
4. Separate traffic-induced events from possible sensor artifacts
5. Use independent labels for damage claims and accuracy metrics such as AUC

Results support unsupervised event prioritization for engineering review. Damage diagnosis and a final model ranking require independent labels and broader testing.

HAROS 2026
Health Monitoring and Retrofitting of Existing Structures

Thank you!

Are there any questions?